

FAIRHOPE 3 NE, AL, USA

WMO: 722370

Lat: 30.5485N

Lon: 87.8757W

Elev: 29

StdP: 100.98

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 63869

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.2	(c) -1.2	(d) -10.2	(e) 1.6	(f) -0.3	(g) -7.3	(h) 2.0	(i) 2.7	(j) 6.6	(k) 18.1	(l) 5.8	(m) 17.6	(n) 0.8	(o) N/A	(p) 0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.8	(c) 33.3	(d) 25.4	(e) 32.4	(f) 25.3	(g) 31.6	(h) 25.2	(i) 27.0	(j) 31.1	(k) 26.6	(l) 30.7	(m) 26.2	(n) 30.2	(o) 1.9	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 25.8	(b) 21.2	(c) 29.3	(d) 25.4	(e) 20.7	(f) 29.0	(g) 25.1	(h) 20.2	(i) 28.7	(j) 84.9	(k) 31.3	(l) 82.9	(m) 30.7	(n) 81.3	(o) 30.2	(p) 28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.7	(b) 4.9	(c) 4.3	(e) -6.7	(f) 35.0	(g) 1.9	(h) 1.4	(i) -8.1	(j) 36.0	(k) -9.2	(l) 36.8	(m) -10.3	(n) 37.6	(o) -11.7	(p) 38.5		
(4) 5.7	(5) 4.9	(6) 4.3	(7) -6.7	(8) 35.0	(9) 1.9	(10) 1.4	(11) -8.1	(12) 36.0	(13) -9.2	(14) 36.8	(15) -10.3	(16) 37.6	(17) -11.7	(18) 38.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.1	9.8	12.2	15.1	18.4	22.5	26.1	26.9	26.9	24.9	19.7	14.0	12.3
	DBStd	7.18	5.68	5.49	4.57	3.44	2.63	1.69	1.42	1.53	2.19	4.21	4.59	5.47
	HDD10.0	179	74	38	11	1	0	0	0	0	0	1	14	40
	HDD18.3	986	268	179	118	40	5	0	0	0	0	36	143	196
	CDD10.0	3504	69	101	169	253	387	484	523	523	447	302	134	113
	CDD18.3	1270	4	9	18	42	133	234	265	264	198	78	13	11
	CDH23.3	11210	4	16	62	272	1144	2308	2614	2497	1598	616	58	22
	CDH26.7	3944	0	0	1	16	298	884	1042	1010	558	132	3	0
Wind	WSAvg	1.6	2.1	2.2	2.1	1.9	1.5	1.2	1.1	1.1	1.3	1.6	1.7	2.1
Precipitation	PrecAvg	1702	140	139	148	116	125	166	208	175	152	90	119	118
	PrecMax	2389	415	365	295	521	349	470	726	437	612	332	306	228
	PrecMin	1041	22	21	10	3	8	18	21	32	14	0	15	42
	PrecStd	325	86	77	70	108	92	104	131	92	119	75	76	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	25.2	26.3	28.3	32.3	34.2	34.1	34.4	33.4	30.9	26.9	25.2
		MCWB	19.9	20.5	19.2	22.0	23.3	25.3	25.9	25.4	24.8	24.3	21.5	22.0
	2%	DB	22.4	23.5	25.1	27.0	30.5	32.7	32.9	33.2	31.8	29.4	24.9	23.9
		MCWB	19.9	19.9	20.0	21.4	22.6	25.0	25.9	25.8	24.4	23.2	20.5	21.0
	5%	DB	21.0	22.0	23.8	26.0	29.4	31.6	32.0	32.3	30.8	28.2	23.7	22.6
		MCWB	18.9	19.3	19.6	20.7	22.8	25.2	25.7	25.6	24.4	22.4	19.9	20.2
	10%	DB	19.5	20.7	22.4	25.0	28.5	30.7	31.2	31.3	30.0	26.8	22.3	21.2
		MCWB	17.6	18.4	18.9	19.9	22.4	25.0	25.4	25.5	24.2	21.6	18.5	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.9	21.9	24.0	25.0	27.3	27.5	27.8	26.7	25.8	23.3	23.0
		MCDB	22.7	23.9	24.1	26.6	28.9	31.5	31.7	32.0	31.1	28.4	25.2	24.5
	2%	WB	20.4	21.0	21.3	23.0	24.4	26.6	26.8	27.1	25.9	24.9	22.2	22.0
		MCDB	21.9	23.0	23.4	25.4	28.3	30.8	31.1	31.2	30.0	27.7	24.1	23.4
	5%	WB	19.3	20.1	20.5	22.4	23.9	26.1	26.4	26.7	25.5	24.0	20.8	20.7
		MCDB	20.7	21.7	22.5	24.7	27.8	30.2	30.7	30.6	29.3	27.0	23.2	22.1
	10%	WB	17.8	19.1	19.7	21.6	23.3	25.6	26.0	26.2	25.0	23.0	19.6	19.5
		MCDB	19.2	20.5	21.7	24.0	27.1	29.5	30.1	29.9	28.6	25.8	21.6	21.0
Mean Daily Temperature Range	5% DB	MDBR	11.3	11.4	11.6	11.6	11.2	9.2	8.8	9.0	9.8	12.0	12.8	10.6
		MCDBR	10.0	9.9	11.0	11.5	12.2	10.2	9.6	10.0	10.5	11.3	12.0	9.9
		MCWBR	7.7	6.4	6.3	6.3	5.7	4.4	4.1	4.4	4.7	6.0	7.9	7.8
	5% WB	MCDBR	8.7	7.6	8.5	9.1	10.0	8.6	8.8	8.9	9.2	9.3	10.2	8.8
		MCWBR	7.5	6.1	6.1	6.2	5.2	4.4	4.1	4.3	4.6	5.9	8.0	7.8
Clear-Sky Solar Irradiance	taub	0.331	0.348	0.372	0.402	0.438	0.463	0.511	0.500	0.445	0.383	0.354	0.340	
	taud	2.546	2.500	2.428	2.345	2.282	2.263	2.149	2.177	2.298	2.456	2.475	2.530	
	Ebn at Noon	902	915	912	892	857	830	790	795	829	866	865	872	
	Edh at Noon	86	98	112	126	135	137	153	147	126	101	91	83	
All-Sky Solar Radiation	RadAvg	3.06	3.62	4.91	5.96	6.59	6.27	5.91	5.54	5.20	4.55	3.48	2.68	
	RadStd	0.24	0.50	0.39	0.47	0.34	0.48	0.41	0.41	0.32	0.46	0.30	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+104

Nomenclature: See separate page